

Computer Science Notes (One-Shot Revision)

1. What is Computer Science?

Definition

Computer Science is the study of computers, computational algorithms, data structures, and software systems. It focuses on solving problems using programming and logical reasoning.

Key Points

- Studies algorithms, data structures, and software systems.
 - Explains how computers process information.
 - Solves real-world problems using programming.
 - Covers:
 - Software Development
 - Artificial Intelligence (AI)
 - Cybersecurity
 - Networking
 - Computational Complexity
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2. What are Data Structures?

Definition

A Data Structure is a way of organizing and storing data so it can be accessed, modified, and managed efficiently.

Common Data Structures

- Array
- Linked List
- Stack
- Queue
- Tree
- Graph

Importance

- Improves program performance.
- Reduces time complexity.
- Optimizes memory usage.
- Essential for efficient algorithm design.

3. What is an Algorithm?

Definition

An algorithm is a step-by-step procedure to solve a problem.

Characteristics

- Takes input.
- Processes through defined steps.
- Produces output.
- Must be:
 - Well-defined
 - Finite
 - Efficient

Performance Measures

- Time Complexity
- Space Complexity

Common Types

- Searching
 - Sorting
 - Recursion
 - Dynamic Programming
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4. What is Time Complexity?

Definition

Time Complexity measures how the execution time of an algorithm grows with the input size.

Examples

- Linear Search → **$O(n)$**
- Binary Search → **$O(\log n)$**

Importance

- Compares algorithm efficiency.
- Helps select the best algorithm.
- Improves scalability and optimization.
- Very important for large datasets.

5. What is Space Complexity?

Definition

Space Complexity is the amount of memory required by an algorithm relative to the input size.

Includes

- Input Space
- Auxiliary Space (extra memory)

Note

- Recursive algorithms require additional stack memory.
 - Measured using **Big-O notation**.
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6. What is Object-Oriented Programming (OOP)?

Definition

Object-Oriented Programming (OOP) is a programming paradigm based on objects that contain data (attributes) and methods (functions).

Four Pillars of OOP

1. Encapsulation

- Hides internal data.
- Protects data from unauthorized access.

2. Abstraction

- Hides implementation details.
- Shows only essential features.
- Simplifies complexity.

3. Inheritance

- One class acquires properties and methods of another.
- Promotes code reuse.

4. Polymorphism

- One interface, multiple implementations.
- Provides flexibility.

Advantages

- Code Reusability
 - Better Security
 - Easy Maintenance
 - Modular Programming
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7. What is a Database?

Definition

A Database is an organized collection of structured data stored electronically.

Database Management System (DBMS)

A DBMS manages databases.

Examples

- MySQL
- PostgreSQL

Advantages

- Efficient data storage.
 - Fast data retrieval.
 - Easy data management.
 - Maintains data integrity.
 - Provides security and consistency.
 - Reduces data redundancy.
 - Supports transactions and constraints.
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8. What is an Operating System (OS)?

Definition

An Operating System is system software that manages computer hardware and software resources and acts as an interface between the user and the hardware.

Responsibilities

- Process Management
- Memory Management
- File System Management
- Device Management

Examples

- Windows
- Linux
- macOS

Importance

- Supports multitasking.
 - Allocates system resources.
 - Provides security.
 - Ensures system stability.
 - Optimizes performance.
 - Enables user-friendly interaction with the computer.
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9. What is a Process and a Thread?

Process

- An independent program in execution.
- Has its own memory space.

Thread

- A lightweight execution unit within a process.
- Shares memory and resources with other threads.

Process vs Thread

Process	Thread
Independent execution	Part of a process
Separate memory	Shared memory
Higher resource usage	Lower resource usage
Slower creation	Faster creation

Key Points

- Processes are isolated.

- Threads communicate easily.
 - Multithreading improves performance.
 - Synchronization is needed to avoid race conditions.
 - Useful for multitasking and high-performance applications.
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10. What is Networking?

Definition

Networking is the process of connecting computers and devices to share data and resources.

Protocol

- TCP/IP defines communication rules.

Types of Networks

- LAN (Local Area Network)
- MAN (Metropolitan Area Network)
- WAN (Wide Area Network)

Applications

- Internet Access
 - File Sharing
 - Cloud Computing
 - Communication Services
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11. Compiler

Definition

- A compiler is a program that translates the entire high-level source code into machine code before execution.

Features

- Checks errors before execution.
- Optimizes the code.
- Improves execution speed.
- Scans the whole program only once.
- Produces an executable file.

Examples

- GCC (C/C++)
 - Java Compiler (javac)
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12. Interpreter

Definition

- An interpreter translates and executes code line by line at runtime.

Features

- No separate compilation step.
- Errors appear only when the specific line executes.
- Easier to debug.
- Slower than compiled programs.

Examples

- Python
 - JavaScript
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Compiler vs Interpreter

Compiler	Interpreter
Translates entire program	Translates line by line
Generates executable file	No executable file
Faster execution	Slower execution
Errors shown after compilation	Errors shown during execution
Better performance	Easier debugging

13. Artificial Intelligence (AI)

Definition

- AI is the field of building systems that perform tasks requiring human intelligence such as learning, reasoning, decision-making, and problem-solving.

Machine Learning (ML)

- A subset of AI.
- Learns patterns from data without explicit programming.

Deep Learning (DL)

- A subset of ML.
- Uses neural networks to learn complex patterns.

AI Systems

Include:

- Data collection
 - Data preprocessing
 - Model training
 - Model evaluation
 - Deployment
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14. Cybersecurity

Definition

- Cybersecurity protects systems, networks, and data from digital attacks.

Security Techniques

- Authentication
- Encryption
- Firewalls

Common Threats

- Malware
- Phishing
- Ransomware

CIA Triad

- Confidentiality
 - Integrity
 - Availability
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15. Cloud Computing

Definition

- Cloud computing provides computing resources over the internet on a pay-as-you-go basis.

Advantages

- No physical infrastructure required.
- Scalable
- Cost-effective

Types of Cloud Services

IaaS (Infrastructure as a Service)

- Provides servers, storage, and networking.
- Example: AWS EC2

PaaS (Platform as a Service)

- Platform to develop and deploy applications.
- Example: Heroku

SaaS (Software as a Service)

- Ready-to-use software over the internet.
 - Example: Gmail
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16. Recursion

Definition

- A technique where a function calls itself until a base case is reached.

Components

- Base Case
- Recursive Case

Problems

- Missing base case causes infinite recursion.
 - Can result in Stack Overflow.
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17. Linked List

Definition

- A linear data structure where elements are stored in nodes connected using pointers.

Advantages

- Dynamic memory allocation.
- No contiguous memory required.
- Easy insertion and deletion.

Applications

- Stack implementation
 - Queue implementation
 - Memory management
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18. Stack

Definition

- A linear data structure that follows the **LIFO (Last In, First Out)** principle.

Operations

- Push
- Pop
- Peek

Applications

- Function calls
- Expression evaluation
- Undo operations

Errors

- Stack Overflow
 - Stack Underflow
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19. Queue

Definition

- A linear data structure that follows the **FIFO (First In, First Out)** principle.

Types

- Simple Queue
- Circular Queue
- Priority Queue

Applications

- CPU Scheduling
 - Buffering
 - Breadth First Search (BFS)
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20. Tree Data Structure

Definition

- A hierarchical data structure consisting of nodes connected by edges.

Components

- Root Node
- Parent Node
- Child Node

Examples

- Binary Tree
- Binary Search Tree (BST)

Applications

- Databases
 - File Systems
 - Searching Algorithms
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21. Graph

Definition

- A non-linear data structure consisting of vertices (nodes) and edges.

Applications

- Social Networks
- GPS Navigation
- Recommendation Systems

Traversal Algorithms

- DFS (Depth First Search)
 - BFS (Breadth First Search)
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22. Software Engineering

Definition

- A systematic approach to designing, developing, testing, and maintaining software systems.

Methodologies

- Agile
- Waterfall

Goals

- Reliability
 - Scalability
 - Maintainability
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23. Machine Learning

Definition

- A subset of AI where systems learn patterns from data and improve performance without explicit programming.

Working

1. Data Collection
2. Data Preprocessing
3. Model Training
4. Model Evaluation
5. Deployment

Types of Machine Learning

Supervised Learning

- Uses labeled data.

Unsupervised Learning

- Finds hidden patterns in unlabeled data.

Reinforcement Learning

- Learns through rewards and penalties.

Applications

- Prediction
 - Recommendation
 - Classification
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24. Big Data

Definition

- Extremely large and complex datasets that traditional tools cannot process efficiently.

3 Vs of Big Data

- Volume
- Velocity
- Variety

Technologies

- Hadoop
 - Apache Spark
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25. Blockchain

Definition

- A distributed ledger technology that securely records transactions across multiple systems.

Features

- Transparency
- Immutability
- Decentralization

Applications

- Cryptocurrency
- Supply Chain Management
- Digital Payments

Topic	One-Line Keyword
1. Computer Science	Computers + Algorithms + Software + Problem Solving
2. Data Structures	Organize data for efficient storage and access
3. Algorithm	Step-by-step solution to solve a problem
4. Time Complexity	Measures execution time using Big-O notation
5. Space Complexity	Measures memory usage using Big-O notation
6. OOP	Encapsulation, Abstraction, Inheritance, Polymorphism (EAIP)
7. Database	Organized data managed by a DBMS
8. Operating System	Manages hardware, software, memory, and processes
9. Process & Thread	Process = Independent execution; Thread = Lightweight execution unit
10. Networking	Connects devices using TCP/IP to share data
11. Compiler	Translates entire source code before execution
12. Interpreter	Executes code line by line during runtime
13. Compiler vs Interpreter	Compiler = Faster; Interpreter = Easier debugging
14. Cybersecurity	Protects systems using the CIA Triad
15. Cloud Computing	On-demand computing resources over the internet
16. Recursion	Function calls itself until a base case
17. Linked List	Dynamic nodes connected using pointers
18. Stack	LIFO – Last In, First Out
19. Queue	FIFO – First In, First Out
20. Tree	Hierarchical data structure with parent-child nodes
21. Graph	Vertices and edges representing relationships
22. Software Engineering	Systematic software development using SDLC methodologies
23. Machine Learning	Learns patterns from data to make predictions
24. Big Data	3Vs: Volume, Velocity, Variety
25. Blockchain	Decentralized, immutable distributed ledger